$$\begin{aligned}\mathcal{C}(\tau) &= \sum_i \frac{c_i^2}{2m_i\omega_i}\left\{\coth\left(\frac{\omega_i}{2k_{\mathrm{B}}T}\right)\cos\left(\omega_i\tau\right) - \mathrm{i}\sin\left(\omega_i\tau\right)\right\} \\ &\equiv \nu(\tau) - \mathrm{i}\eta(\tau). \end{aligned} \tag{5.15}$$